

Issue # 7: Numerator Data - Descriptive Statistics 2023-2024

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This Write-up contains tables and plots for the descriptive analysis of the Numerator data. The original user-trip-item level data is transformed to user-month level data, with variables summarized at the monthly level for each user.

1. Tables

- FAH/FAFH department codes (less granular grouping) and soda category (more granular grouping) codes
- Number of users by state-year
- Demographics by treatment status
- Spending by treatment status
- Soda spending by treatment status
- Food and soda spending percentiles

2. Figures: Distribution of ...

- proportion of food spending out of total spending
- proportion of FAH/FAFH spending out of total food spending
- proportion of FAH/FAFH on SNAP out of SNAP spending
- soda spending/winsorized spending
- Proportion of soda spending out of total food spending
- proportion of soda on SNAP out of total SNAP spending
- proportion soda at home/away from home

Major Changes: included online receipts; updated FAH and FAFH classification; relabelled outcome being plotted and updated footnotes; added summary statistics by SNAP status.

Major Observation: Even after adding online receipts that could potentially include food away from home (delivery foods), FAFH is still a very tiny portion. The proportion of total food spending on FAFH actually decreases when comparing to the one with only in-store receipts, potentially suggesting that most of online receipts are also categorized as FAH instead of FAFH.

In summary tables by SNAP treatment status, those who are on SNAP tend to spend more in total and on food.

All these could be a result of Numerator’s financial incentive for uploading receipt that varies based on receipt characteristics (grocery or restaurant) and values. According to the paper by Song, financial incentive increases with receipt values; every grocery receipt receives financial incentive but the restaurant (QSR) receipts will be put on lottery so not all food away from home receipts get financial reward. This can create selection of receipt and potentially results in missing receipts.

Table 1: Food at Home vs. Food Away from Home Categories

Food At Home (Grocery)	Food Away From Home (QSR)
Snack	LSR Sandwiches & Wraps
Canned	LSR Beverages
Produce	LSR Breakfast
Dairy	LSR Desserts
Beverages	LSR Entrees
Packaged Bakery (Bread & Alternatives)	LSR Snack & Sides
Condiments	LSR Mexican
Frozen Foods	LSR Sauces & Condiments
Shelf Stable Meals	LSR Italian
Candy (Snacks)	LSR Salads
Baking & Cooking	
Meat	
Seafood & Fish	
Beans & Grains	
Refrigerated Foods	
Breakfast	
Pasta & Noodles	
Deli & Prepared Foods	
In-Store Bakery (Bread & Alternatives)	
Herbs & Spices	
Alcohol Beverages	
Misc. Food	
Fresh Meal Kits	
Ice	

Table 2: Soda Categories

Soda At Home (Grocery Beverages)	Soda Away From Home (QSR Beverages)
Soft Drinks	LSR Bottled Soda
Tonic & Club Soda	LSR Fountain Drink
Sparkling Juice & Ciders	
Prebiotic & Probiotic Soda	

Table 3: Number of Users in Affected and Unaffected States

State	2023	2024
Household affected in Jan 2026		
FL	26817	26959
IA	3531	3539
ID	1681	1684
IN	8404	8416
LA	4722	4727
NE	2138	2145
OK	3774	3786
UT	2543	2544
WV	2421	2424
Subtotal	56031	56224
Household affected after Jan 2026		
AK	626	628
CO	4586	4609
TX	27632	27763
Subtotal	32844	33000
Treated States	88875	89224
Control States	241707	242565
Total	330582	331789

Notes: In static panel, each static period is defined by a 12-month window, and the panel is balanced within each static period but may not be balanced between static period. Here, we extract the purchases of households who have ever being included in the static panel.

Table 4: User-Month Panel - Demographics

	Treated States	Mean (SD) Control States	Total	Min	Max
Income	78836.50 (58273.74)	83967.38 (62399.32)	83099.17 (61750.58)	10000.00	275000.00
Household Size	2.79 (1.45)	2.86 (1.48)	2.84 (1.47)	1.00	7.00
Age	50.28 (13.07)	48.82 (12.85)	49.06 (12.90)	19.00	67.50
Married	0.57 (0.50)	0.54 (0.50)	0.55 (0.50)	0.00	1.00
Education (years)	13.77 (1.35)	13.82 (1.38)	13.81 (1.38)	10.00	16.00
White	0.74 (0.44)	0.69 (0.46)	0.70 (0.46)	0.00	1.00
Black	0.09 (0.28)	0.10 (0.30)	0.10 (0.30)	0.00	1.00
Asian	0.03 (0.16)	0.07 (0.26)	0.07 (0.25)	0.00	1.00
Hispanic	0.13 (0.33)	0.12 (0.32)	0.12 (0.32)	0.00	1.00
Ever Child	0.23 (0.42)	0.24 (0.43)	0.24 (0.43)	0.00	1.00
Number of States	9	48	57		
Number of Users	56,316	275,954	332,270		
Average Number of Months per User	23.23	23.28	23.27		

Notes: Income's 'less than 20000' category is coded as 10000, and 'more than 250000' is coded as 275000. Other income buckets are coded as the mean of a bucket. Education's 'advanced' category is coded as 16, and 'less than high school' is coded as 10.

Table 5: User-Month Panel - Spending

	Treated States	Mean (SD) Control States	Total	Min	Max
Food Spending	549.68 (451.33)	547.50 (442.72)	547.87 (444.19)	0.00	50429.40
Total Spending	1367.41 (1124.72)	1358.20 (1108.91)	1359.76 (1111.61)	0.00	135684.55
Prop. of Total Spending on Food	0.43 (0.20)	0.43 (0.20)	0.43 (0.20)	0.00	1.00
Prop. of Food Spending on FAH	0.90 (0.14)	0.90 (0.15)	0.90 (0.15)	0.00	1.00
Prop. of Food Spending on FAFH	0.10 (0.14)	0.10 (0.15)	0.10 (0.15)	0.00	1.00
Prop of Food on EBT	0.33 (0.27)	0.33 (0.27)	0.33 (0.27)	0.00	1.00
Prop. of EBT Spending on FAH	1.00 (0.02)	1.00 (0.03)	1.00 (0.03)	0.00	1.00
Prop. of EBT on FAFH	0.00 (0.02)	0.00 (0.03)	0.00 (0.03)	0.00	1.00
Ever SNAP	0.16 (0.37)	0.17 (0.38)	0.17 (0.37)	0.00	1.00
Number of States	9	48	57		
Number of Users	56,316	275,954	332,270		

Notes: Prop. of food spending is calculated out of total spending. Prop. of FAH/FAFH spending and food on SNAP is calculated out of total food spending. Prop. of FAH/FAFH on SNAP is calculated out of total SNAP spending. Means are calculated by averaging users' monthly spending/proportion, pooled over time and users. No winsorization. EBT-related outcomes are conditional on households on SNAP.

Table 6: User-Month Panel - Soda Spending

	Treated States	Mean (SD) Control States	Total	Min	Max
Soda Spending	22.70 (35.07)	21.21 (35.28)	21.46 (35.25)	0.00	11350.8
Prop. of Food Spending on Soda	0.04 (0.06)	0.04 (0.06)	0.04 (0.06)	0.00	1.00
Prop. of Soda Spending on Soda at Home	0.88 (0.25)	0.87 (0.27)	0.87 (0.26)	0.00	1.00
Prop. of Soda Spending on Soda Away From Home	0.12 (0.25)	0.13 (0.27)	0.13 (0.26)	0.00	1.00
Prop of EBT Spending on Soda	0.06 (0.12)	0.06 (0.12)	0.06 (0.12)	0.00	1.00
Prop of Soda Spending on EBT	0.30 (0.35)	0.30 (0.36)	0.30 (0.36)	0.00	1.00
Number of Users	56,316	275,954	332,270		

Notes: Prop. of soda spending is calculated out of total food spending. Prop. of soda FAH/FAFH spending is calculated out of total food spending. Prop. of soda on SNAP is calculated out of total SNAP spending. Means are calculated by averaging users' monthly spending/proportion, pooled over time and users. No winsorization.

Table 7: User-Month Panel - Demographics (SNAP vs Non-SNAP)

	SNAP	Mean (SD) Non-SNAP	Total	Min	Max
Income	46740.95 (47459.10)	89451.63 (61761.73)	83099.17 (61750.58)	10000.00	275000.00
Household Size	3.02 (1.68)	2.81 (1.43)	2.84 (1.47)	1.00	7.00
Age	48.36 (12.80)	49.19 (12.91)	49.06 (12.90)	19.00	67.50
Married	0.31 (0.46)	0.59 (0.49)	0.55 (0.50)	0.00	1.00
Education (years)	13.31 (1.43)	13.90 (1.35)	13.81 (1.38)	10.00	16.00
White	0.54 (0.50)	0.72 (0.45)	0.70 (0.46)	0.00	1.00
Black	0.19 (0.39)	0.08 (0.28)	0.10 (0.30)	0.00	1.00
Asian	0.05 (0.22)	0.07 (0.25)	0.07 (0.25)	0.00	1.00
Hispanic	0.19 (0.39)	0.11 (0.31)	0.12 (0.32)	0.00	1.00
Ever Child	0.29 (0.46)	0.23 (0.42)	0.24 (0.43)	0.00	1.00
Number of States	54	57	57		
Number of Users	49,814	282,456	332,270		
Average Number of Months per User	23.09	23.30	23.27		

Notes: Income's 'less than 20000' category is coded as 10000, and 'more than 250000' is coded as 275000. Other income buckets are coded as the mean of a bucket. Education's 'advanced' category is coded as 16, and 'less than high school' is coded as 10.

Table 8: User-Month Panel - Spending (SNAP vs Non-SNAP)

	SNAP	Mean (SD) Non-SNAP	Total	Min	Max
Food Spending	636.22 (500.30)	532.44 (431.79)	547.87 (444.19)	0.00	50429.40
Total Spending	1409.99 (1124.42)	1350.98 (1109.12)	1359.76 (1111.61)	0.00	135684.55
Prop. of Total Spending on Food	0.48 (0.20)	0.42 (0.20)	0.43 (0.20)	0.00	1.00
Prop. of Food Spending on FAH	0.90 (0.13)	0.90 (0.15)	0.90 (0.15)	0.00	1.00
Prop. of Food Spending on FAFH	0.10 (0.13)	0.10 (0.15)	0.10 (0.15)	0.00	1.00
Prop of Food on EBT	0.38 (0.27)	0.21 (0.23)	0.33 (0.27)	0.00	1.00
Prop. of EBT Spending on FAH	1.00 (0.03)	1.00 (0.04)	1.00 (0.03)	0.00	1.00
Prop. of EBT on FAFH	0.00 (0.03)	0.00 (0.04)	0.00 (0.03)	0.00	1.00
Ever SNAP	0.79 (0.41)	0.06 (0.24)	0.17 (0.37)	0.00	1.00
Number of States	54	57	57		
Number of Users	49,814	282,456	332,270		

Notes: Prop. of food spending is calculated out of total spending. Prop. of FAH/FAFH spending and food on SNAP is calculated out of total food spending. Prop. of FAH/FAFH on SNAP is calculated out of total SNAP spending. Means are calculated by averaging users' monthly spending/proportion, pooled over time and users. No winsorization.

Table 9: User-Month Panel - Soda Spending (SNAP vs Non-SNAP)

	SNAP	Mean (SD) Non-SNAP	Total	Min	Max
Soda Spending	29.55 (40.60)	20.05 (34.03)	21.46 (35.25)	0.00	11350.87
Prop. of Food Spending on Soda	0.05 (0.06)	0.04 (0.06)	0.04 (0.06)	0.00	1.00
Prop. of Soda Spending on Soda at Home	0.90 (0.23)	0.87 (0.27)	0.87 (0.26)	0.00	1.00
Prop. of Soda Spending on Soda Away From Home	0.10 (0.23)	0.13 (0.27)	0.13 (0.26)	0.00	1.00
Prop of EBT Spending on Soda	0.05 (0.11)	0.06 (0.15)	0.06 (0.12)	0.00	1.00
Prop of Soda Spending on EBT	0.35 (0.37)	0.18 (0.30)	0.30 (0.36)	0.00	1.00
Number of Users	49,814	282,456	332,270		

Notes: Prop. of soda spending is calculated out of total food spending. Prop. of soda FAH/FAFH spending is calculated out of total food spending. Prop. of soda on SNAP is calculated out of total SNAP spending. Means are calculated by averaging users' monthly spending/proportion, pooled over time and users. No winsorization.

Table 10: User-Month Panel - Food Spending Percentiles

percentile	food_spending_2023	food_spending_2024
50	450.25	485.90
60	551.52	585.58
70	669.14	702.03
80	821.28	852.90
90	1062.97	1092.84
95	1300.38	1328.24
96	1377.92	1406.04
97	1480.44	1507.77
98	1632.05	1656.17
99	1916.38	1937.33
99.5	2264.36	2278.74
100	40382.87	50429.40

Table 11: User-Month Panel - Soda Spending Percentiles

percentile	soda_spending_2023	soda_spending_2024
50	8.24	10.00
60	14.27	16.45
70	22.00	24.70
80	34.03	37.38
90	56.27	60.55
95	80.70	86.12
96	89.03	94.81
97	99.96	106.22
98	116.25	123.05
99	145.95	153.77
99.5	178.86	188.00
100	11350.87	4843.32

Figure 1: Proportion of Food Spending

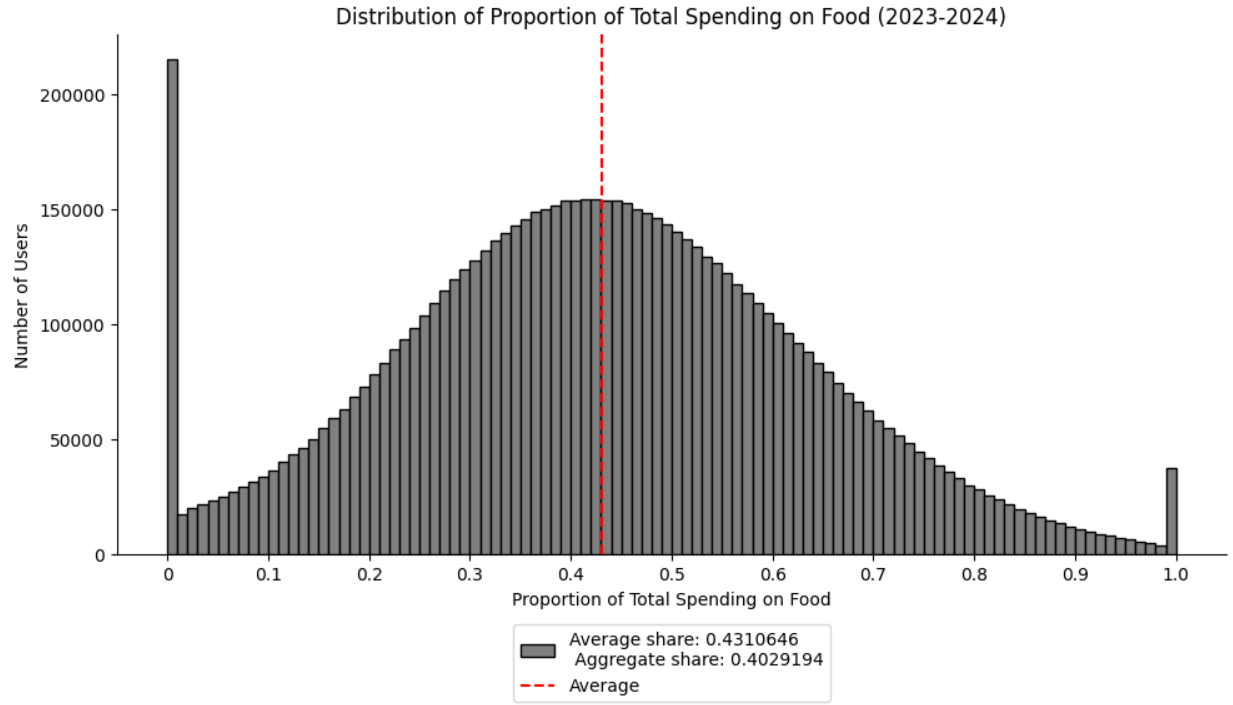


Figure 2: Proportion of FAH/FAFH Spending out of total food spending

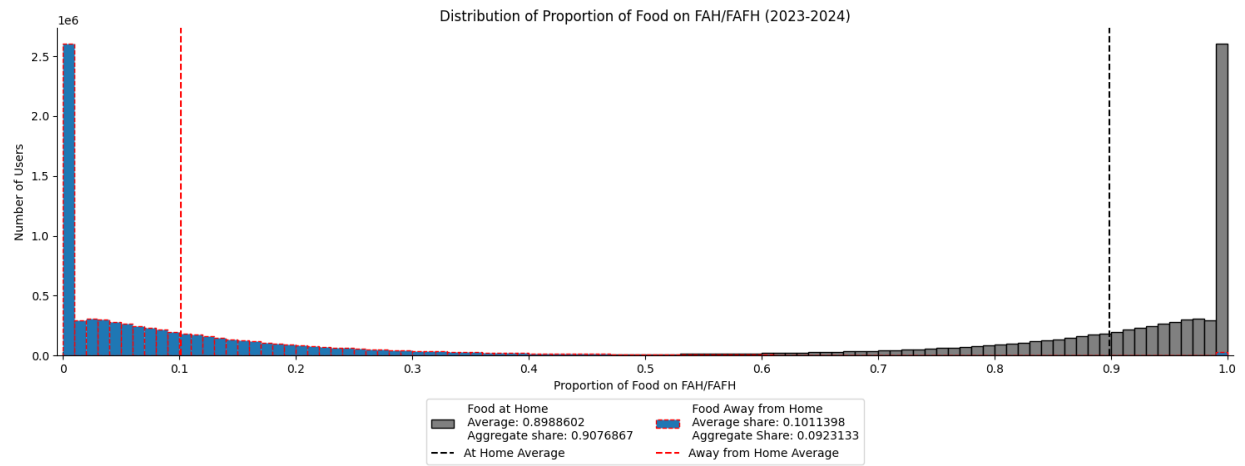
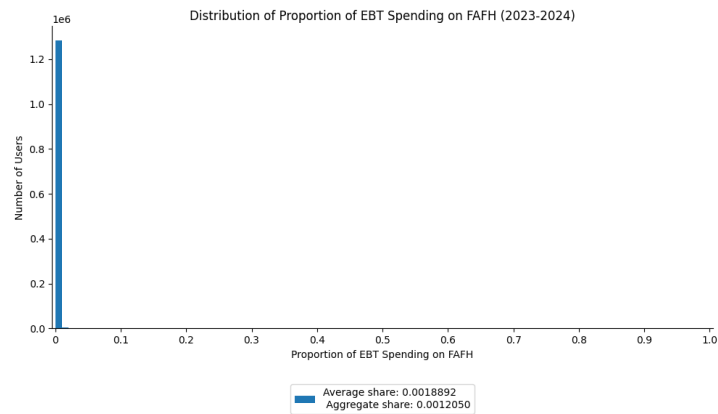


Figure 3: Proportion of FAH/FAFH SNAP Spending out of total SNAP spending

(a) Food Away



(b) Food at Home

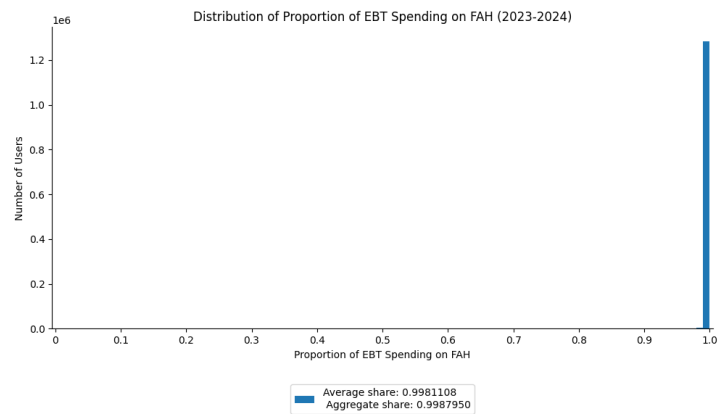
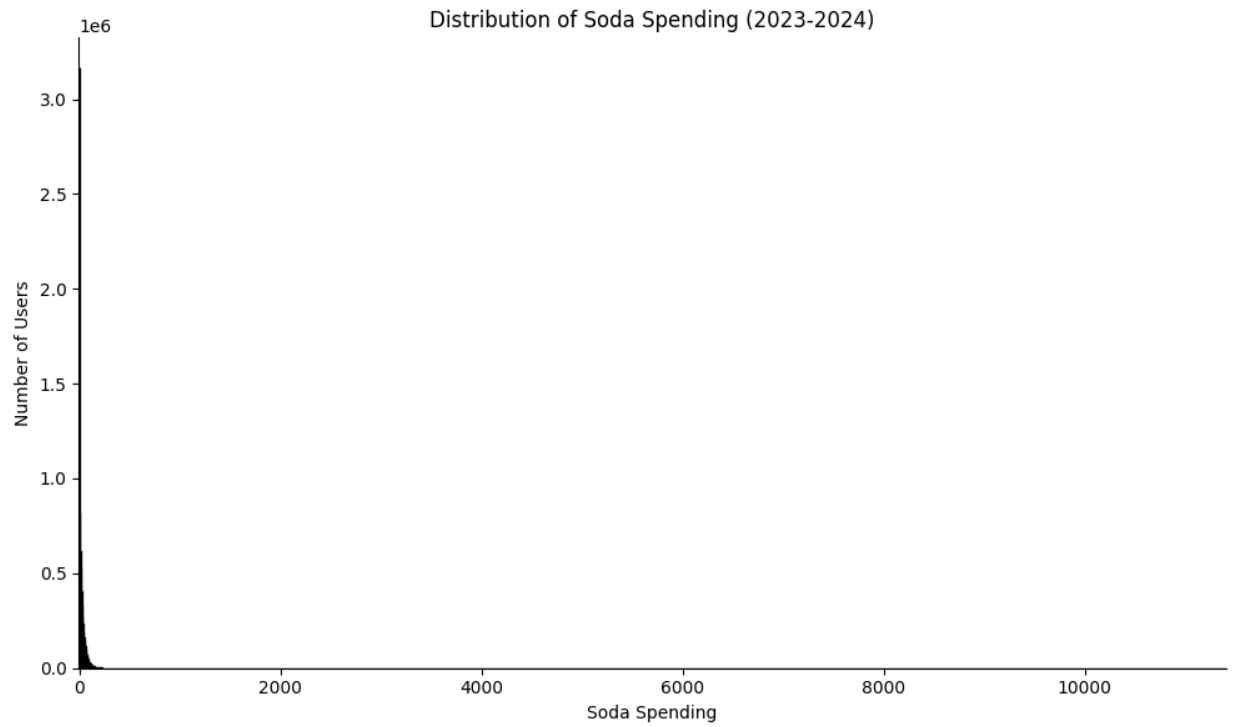


Figure 4: Soda Spending

(a) unwinsorized



(b) winsorized at 99.5%

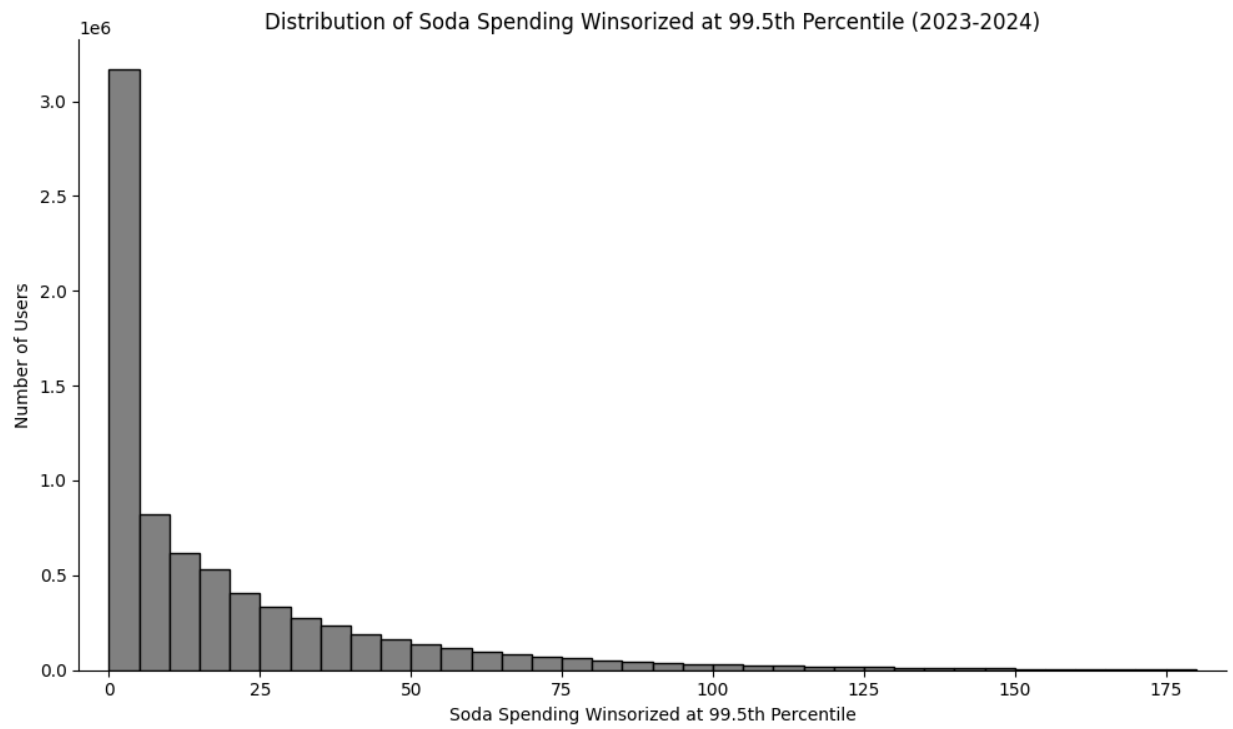


Figure 5: Proportion of Soda Spending out of total food spending

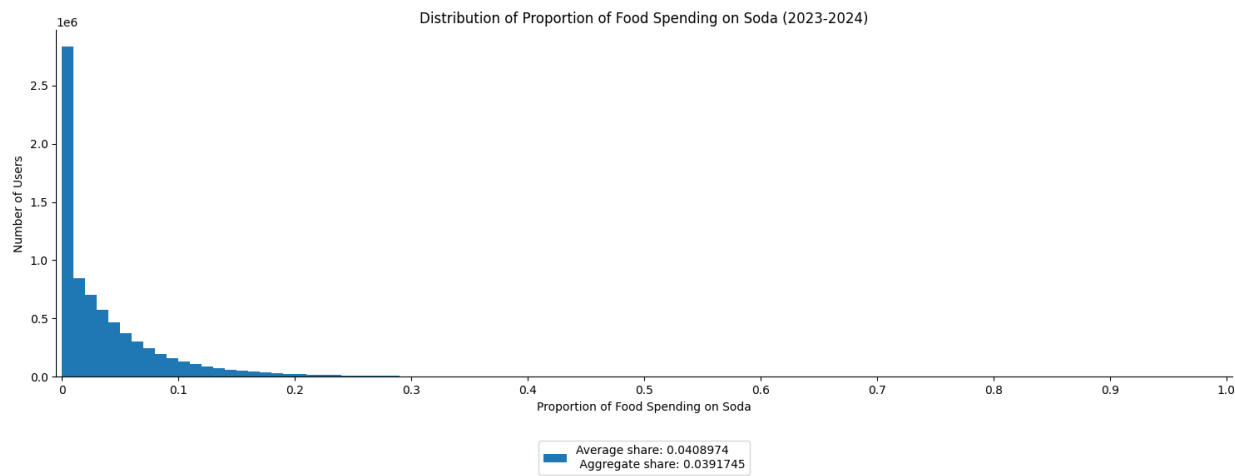


Figure 6: Proportion of Soda SNAP Spending out of total SNAP spending

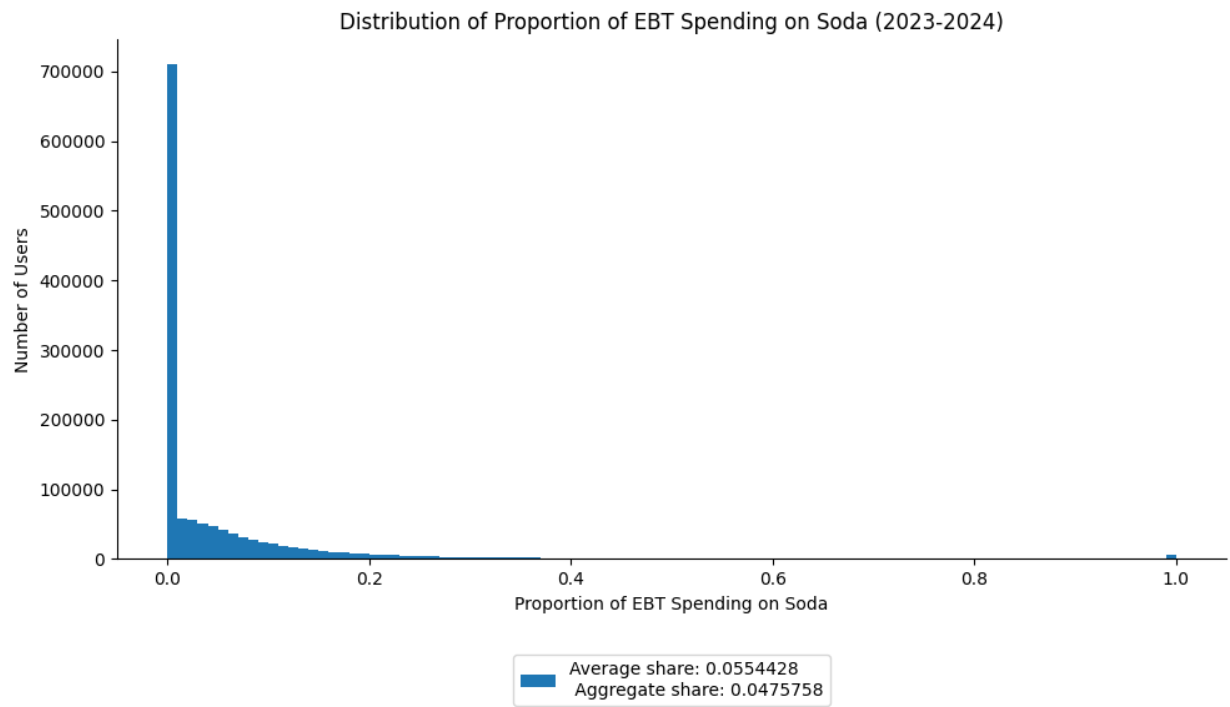


Figure 7: Proportion of Soda FAH/FAFH Spending out of total soda spending

